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Optimal Sortition: A Framework Analysis

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A thesis submitted for the degree of
Bachelor of Commerce (Honours)
at Monash University in 2026
Department of Econometrics & Business Statistics

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Abstract

Sortition, the random selection of individuals to form panels, has seen a sort of renaissance in recent years, as more and more institutions realise the value of direct community and stakeholder input when making decisions. These panels have proven to be massively beneficial in terms of improving the quality of decisions, increasing public trust and engagement, and creating more inclusive and representative decision-making processes. However, there are still many challenges to overcome in order to ensure that these panels are truly representative and effective. This thesis aims to evaluate current sortition methods and community representation processes, as well as to investigate how unobserved traits and survey non-response may affect panel representativeness. By investigating these factors, we can develop a greater understanding of existing tools and procedures, how they work, how they fall short, and what it would mean for them to be more representative, which will then enable the creation of a diagnostic tool to gauge effectiveness.

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The material in Chapter 1 has been submitted to the journal *Journal of Impossible Results* for possible publication.

The contribution in Chapter 2 of this thesis was presented in the International Symposium on Nonsense held in Dublin, Ireland, in July 2022.

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Chapter 1

Introduction

1.1 Aim

This study aims to evaluate current sortition methods and community representation processes, as well as to investigate how unobserved traits and survey non-response may affect panel representativeness. By investigating these factors, we can develop a greater understanding of existing tools and procedures, how they work, how they fall short, what it would mean for them to be more representative, which will then enable the creation of a diagnostic tool to gauge effectiveness.

1.2 Background

We live in an age where decision-making and its impacts feel like they are only growing in importance and scale, yet our ability to impart influence seems to only be diminishing. Many feel disenfranchised and like power is being taken away from them, and that their voices are not being heard. This is a problem. Not only because it can lead to disengagement, apathy, and a lack of trust in institutions, but also because *not* listening directly to lived experiences and perspectives limits the effectiveness of the very decisions we *do* make.

According to Flanigan et al. (2021), sortition creates groups that are statistical "mini-publics" of society, reflecting demographics and perspectives more accurately than elected bodies. Which, given the polarised and over-dramatised nature of policy today, is a much-needed breath of fresh air.

Sortition is a method of selection that uses random sampling to create panels that are representative of the population. It has been used in various contexts, reaching as far back as ancient Athens, where it was how jurors were selected, and into modern times for citizens' assemblies. The idea is that by randomly selecting individuals from the population, we can create panels that are more representative

and inclusive than those selected through traditional methods. Sortition also reduces dominance by elites or highly engaged groups, enabling broader participation across social classes (Giuliani 2025), ensuring decisions actually serve the people they are meant to.

An increasing number of institutions (including the Parisian Citizens' Assembly, the Irish Citizens' Assembly, and the British Columbia Citizens' Assembly) are using sortition to create panels that are more representative of the population. However, there are still many challenges to overcome in order to ensure that these panels are truly representative and effective.

One of the main challenges is ensuring that the selection process is transparent and fair, and that the panels are truly representative of the population in terms of demographics and perspectives. But this raises the questions of both a) what does it mean to be fair? and b) what does it mean to be representative?

Additionally, though some (such as LexiMin, from Flanigan et al. (2021)) have already been developed, there is a need for better tools, methods and frameworks to evaluate the representativeness of these panels and to ensure that they are robust in achieving their goals.

Chapter 2

Literature Review

2.1 Sortition and Representation

2.2 LexiMin Panel Selection (Flanigan et al. 2021)

2.2.1 Problem Definition

Suppose there is a pool of individuals numbered 1 to n.

We want to choose a panel of size k.

Let K be the set of all feasible panels. A feasible panel is any group of k individuals that satisfies the required constraints (which generally take the form of quotas).

2.2.2 LexiMin Approach

LexiMin does not choose one panel immediately. Instead, it chooses a probability distribution over all feasible panels.

Each feasible panel is assigned a probability.

These probabilities must be non-negative and must add up to 1.

For each individual, we then calculate their overall probability of being selected by adding up the probabilities of all feasible panels that include that individual.

Next, we sort the individual selection probabilities from smallest to largest.

The LexiMin rule works by first making the smallest selection probability as large as possible. Subject to that, it makes the second-smallest as large as possible. It then does the same for the third-smallest, and continues in this way.

2.2.3 Formal Representation

Let p_S be the probability assigned to panel S.

These probabilities satisfy:

$$p_S \geq 0$$

and

$$\sum_{S \in K} p_S = 1$$

The selection probability for individual i is:

$$\pi_i(p) = \sum_{S \in K,\ i \in S} p_S$$

This gives the vector of individual selection probabilities:

$$\pi(p) = (\pi_1(p), \pi_2(p), ..., \pi_n(p))$$

Sort these from smallest to largest:

$$\pi_{(1)}(p) \leq \pi_{(2)}(p) \leq ... \leq \pi_{(n)}(p)$$

LexiMin chooses p so that this sorted vector is maximised in order.

2.2.4 Algorithm

1. List all feasible panels in K.
2. Consider all possible probability distributions over those panels.
3. For each distribution, compute:

$$\pi_i(p) = \sum_{S \in K,\ i \in S} p_S$$

4. Sort the selection probabilities:

$$\pi_{(1)}(p) \leq \pi_{(2)}(p) \leq ... \leq \pi_{(n)}(p)$$

5. Choose the distribution that maximises the smallest selection probability.
6. If more than one distribution still remains, choose the one that maximises the second-smallest selection probability.
7. Continue in this way until the distribution is determined.
8. Sample a final panel according to that distribution.

2.2.5 Logic

LexiMin is a fairness rule. It gives priority to the individuals with the lowest chance of being selected and improves their chances first. Then it improves the next-lowest probabilities, and so on.

This makes the selection probabilities as equal as possible while still respecting the panel constraints.

2.2.6 Summary

LexiMin chooses a probability distribution over feasible panels that makes individual selection probabilities as equal as possible, starting with the lowest.

2.3 Visualisation Methods

(VanderPlas et al. 2023)

2.3.1 Parallel coordinate plots

- Can help to show the distribution of traits across the selected panel and how it compares to the overall population.
- Can also show the trade-offs between different traits and how they are balanced in the selection process
- Can be used to identify any segments of the population that are underrepresented in the selected panel, and to highlight any potential biases in the selection process.

(Conway, Lex & Gehlenborg 2017)

2.3.2 Matrix-based set representation (UpSetR)

- Can be used to show how different traits connect with each other and how they are distributed across the distribution and/or selected panel.

- Can help to identify any clusters of traits that are overrepresented or underrepresented in the selected panel, and to show how different traits interact with each other in the selection process.
- Can highlight specific combinations of traits that are of interest, and show how they are represented in the selected panel compared to the overall population. (Though there is no way to gauge the prevalence of these combinations in the population, beyond anecdotal observations)

2.3.3 Query-based highlighting

- Makes the visualisation interactive, allowing users to explore the data and identify specific patterns or trends in the selection process.
- Can be used to highlight specific traits or combinations of traits that are of interest, and to show how they are represented in the selected panel compared to the overall population.
- Also creates opportunity to ensure specific combinations of traits are represented in the selected panel, without reintroducing representative burden.

2.3.4 Attribute distribution analysis

- Showcase how the distribution of traits in the selected panel compares to the overall population, and to identify any discrepancies or biases in the selection process.

2.4 Working questions

2.4.1 Sortition

- For the new quota tool, would it just be a matter of making a concatenated trait quota, evaluating whether the resulting panel is still feasible and meets the requirements of the original quotas, and then tested to see whether it majorly impacted selection likelihood, vs if it were not to have been used?

2.4.2 Visualisation

- What do we actually want to show with these visualisations? What are the key insights we want to communicate?
- Are we looking at likelihood of selection across traits/through stages? Or are we looking at the distribution of traits in the final panel?

Chapter 3

Research Question

What does it mean for a panel to be “good” at being representative and are we achieving this? - Are current methods and algorithms, like LexiMin, performing the best they could be? - How do unobserved/unfiltered traits affect response rates, and therefore panel composition and representativeness? - How can survey non-response and other sampling-level issues bias representativeness?

nDF suggests 10x response to panel count; not generally feasibly, more often get 3x. How much of a difference does this make to burden and selection likelihood?

Chapter 4

Research Plan

Plan

Chapter 5

Preliminary Results

This chapter contains a summary of the context in which your research is set.

Bibliography

Conway, JR, A Lex & N Gehlenborg (2017). UpSetR: An R Package for the Visualization of Intersecting Sets and Their Properties. *Bioinformatics* **33**(18). Ed. by J Hancock, 2938–2940. (Visited on 04/09/2026).

Flanigan, B, P Gölz, A Gupta, B Hennig & AD Procaccia (2021). Fair algorithms for selecting citizens' assemblies. *Nature* **596**, 548–552.

Giuliani, A (2025). Sortition in Politics: From History to Contemporary Democracy. *Parliaments, Estates and Representation* **45**(3), 317–335.

VanderPlas, S, Y Ge, A Unwin & H Hofmann (2023). Penguins Go Parallel: A Grammar of Graphics Framework for Generalized Parallel Coordinate Plots. *Journal of Computational and Graphical Statistics* **32**(4), 1572–1587. (Visited on 04/09/2026).